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When learning from limited experience takes off: Linear/curvilinear relationships between decision-making comprehensiveness and sustainable supply chain performance under contingent conditions

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ABSTRACT

Studies examining the effect of sustainability decision-making comprehensiveness among supply chain alliances on sustainable supply chain management (SSCM) performance have been underrepresented in the current literature. This linking relationship is investigated herein through the theoretical lens of the Organizational Learning Curve (OLC) and Learning from Limited Experience (LLE), and detailed insights under contingent conditions are also provided. The field research is based on a mixed-method approach, including two sequential studies—a quantitative survey followed by a qualitative study in Vietnam. Their results confirm and validate the proposed *J*-shaped, curvilinear effects of sustainability decision-making comprehensiveness on economic and social sustainability performances and a linear effect on environmental sustainability performance. Moreover, competitive intensity and opportunism moderate these direct effect relationships. We contribute to a crucial gap within the SSCM literature and expand the applications of both OLC and LLE while drawing up a set of practical guidelines in the studied subject matter.

1. Introduction

Due to the emergence of social big data, wherein massive and diverse amounts of data are generated in real-time via online interactions between a number of entities, group decision-making involving multiple participants has become viable in practice, and substantial scholarly focus has been garnered on this topic in the field of decision science [1–4]. The concept of group decision-making transcends individual teams/firms and has evolved to encompass inter-organizational levels characterized by a broader scale and varying backgrounds, not constrained by time and/or location [ibid]. For sustainable supply chain management (SSCM hereinafter), in particular, decision-making involves evaluating and selecting alternatives that not only reduce costs and optimize efficiency but also minimize adverse impacts on the environment, promote ethical labor practices, and align with social responsibility benchmarks within the supply chain system [5]. Such decision-making can be complex and multifaceted, requiring a holistic approach to ensure decision quality (see, [6]). These factors underscore the necessity of group decision-making among supply chain alliances to pool resources, expertise, and data so as to enable a more comprehensive and well-informed approach to addressing sustainability challenges [7]. Numerous firms have adopted group decision support systems to improve daily sustainable supply chain operations for activities such as data aggregation, modeling, forecasting, and planning [8]. Nike, for example, employs collaborative decision-making involving stakeholders, suppliers, and customers [9]. Walmart utilizes a comparable approach in its SSCM, collaborating with suppliers, distributors, and logistics providers to tackle sustainability concerns including energy efficiency, carbon emissions, etc. [10]. Group decision-making in supply chain management or SSCM has emerged as a trending focus of research

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in recent years (see [11]).

Since sustainability has been globally acknowledged and emphasized as critical for long-term viability, the fundamental goal of group decision-making in supply chains is to achieve comprehensive and high-quality decisions that lead to improved SSCM performance (cf., [7]). However, the practical effect of the increasing comprehensiveness level in such group decision-making on SSCM performance is still shrouded in uncertainty. While the influence of decision-making comprehensiveness on organizational outcomes in general has been well-established (e.g., [12]), upon a meticulous scrutiny of the literature on comprehensive group decision-making, distinct research gaps come to light and are outlined in the discussion that follows. More comprehensive discussions are provided in the Literature and Hypotheses section, with Table 1 listing the related literature for greater clarity.

First, a significant inconsistency arises regarding the effects of decision-making comprehensiveness on organizational performance outcomes. Despite extensive efforts to uncover these effects under varying contextual factors, a clear consensus remains elusive. Fredrickson and Mitchell [13] reveal in their pioneering work a negative relationship between comprehensiveness and return on assets/sales growth in a volatile industry. This viewpoint is echoed by the study of Souitaris and Maestro [14], as it affirms the adverse impact of decision comprehensiveness on financial performance in dynamic and uncertain environments. In contrast with these contrary relationships, another perspective suggests the effectiveness of comprehensive decisions in various contexts such as uncertainty and instability (e.g., [15,16]). Moreover, some scholars have proposed nonlinear effects in decision comprehensiveness (e.g., [17]). Notably, the empirical results of an in-depth analysis by Miller [ibid], a renowned researcher investigating decision comprehensiveness, highlight inconsistencies in specific performance outcomes. These divergent findings indicate the dynamic nature of decision comprehensiveness effects, contingent on contextual intricacies and performance categories [18].

Second, research on decision-making comprehensiveness has been conducted at various levels, including individual [19,20], team/group [14,21], and inter-organizational [22]. However, as pointed out by an earlier work of Walter et al. in 2012 [ibid], a significant gap exists in understanding inter-organization group decision-making. A review of the literature spanning the last decade has also revealed a paucity of such studies, in particular regarding the effects of such decision-making on firm performance (see for example, [5,23]). This, in turn, has limited a thorough understanding of group decision-making processes at the inter-organizational level, thus hindering a deeper grasp of intricate interfirm dynamics.

Lastly, a significant research gap exists in understanding decision-making comprehensiveness in SSCM. While the significance of sustainability is recognized, recent studies have begun discussing the impact of comprehensive decision-making on SSCM performance. However, these studies mainly propose conceptual frameworks (e.g., [5]) without empirically and quantitatively assessing the effect of comprehensive decision-making on SSCM outcomes. Thus, a research opportunity exists in empirically investigating the impacts of comprehensive decision-making in SSCM. As a result, a notable research question arises concerning the differentiated impacts of comprehensiveness in group decision-making at the inter-organizational level on various SSCM performance outcomes, including economic, social, and environmental performance, thus yielding answers that have previously been inconclusive.

In response to the research inquiry noted above, this study develops a theoretical framework informed by established theories and empirical data. Building upon the role of knowledge and experience in decision-making outcomes (e.g., [17]), the Learning from Limited Experience theory (LLE hereinafter) [24], which is adapted from the Organizational Learning Curve theory (OLC hereinafter) [25], is employed to develop the framework. The study anticipates a curvilinear effect of comprehensive decision-making in the supply chain system on economic and social sustainability performances, while environmental performance follows a linear pattern as per insights from the OLC theory. Furthermore, while most initial literature often conflates ambiguity, uncertainty, and instability when predicting the moderating effect of decision comprehensiveness (e.g., [18]), Forbes [ibid] suggests that an organizational information environment is better suited to contingently capture managerial decision contexts, thus deriving various values of decision comprehensiveness. Therefore, this study introduces

Table 1
A literature review of decision-making.

no	Article	Group decision-making at the inter-organiza-tional level	Effect of decision-making comprehen-siveness	Sustainable supply chain management	Research purpose	Metho-dology
1	Miller (2008)		X		Examining the effect of decision-making comprehensiveness on performance (including profitability and effectiveness)	Survey
2	Schrettle et al. (2014)		X	X	Proposing the effect of decision-making comprehensiveness on SSCM performance	Conceptual framework
3	Walter et al. (2012)	X	X		Examining the effect of group decision-making comprehensiveness on general performance	Survey
4	Zwikael & Meredith, (2019)	X	X		Examining the effect of group decision-making comprehensiveness on investment performance	Survey
5	Allaoui et al. (2019)	X		X	Proposing methods to improve group decision-making comprehensiveness in SSCM	Conceptual framework
6	Salmela & Huiskonen (2019)	X		X	Proposing methods to improve group decision-making comprehensiveness in SSCM	Case study
7	Liu & Bao (2021)	X		X	Proposing methods to improve group decision-making comprehensiveness in SSCM	Fuzzy TOPSIS method
8	Guo et al. (2022)	X		X	Proposing methods to improve group decision-making comprehensiveness in SSCM	Mixed-integer linear programming model
9	He & Xiong (2022)	X		X	Proposing methods to improve group decision-making comprehensiveness in SSCM	Simulation
10	Curşeu & Schruijer (2017)	X	X	X	Proposing the effect of group decision-making comprehensiveness on SSCM performance	Conceptual framework
This study		**X**	**X**	**X**	**Examining the effect of group decision-making comprehensiveness on SSCM performance**	**Survey**

Source: this research; a full reference list will be provided upon request.

competitive intensity and opportunism as specific conditions to illuminate the interaction effect of the information environment and organizational decisions on performance. Competitive intensity necessitates real-time data for informed decisions [26], while opportunism, driven by self-interest, can distort/manipulate information [27].

With the above intentions, this study endeavors to address the following key questions: 1) Does sustainability decision-making comprehensiveness trigger SSCM performance outcomes, including economic, social, and environmental? 2) What type of causal relationship exists between the two variables? and 3) Do competitive intensity and opportunism act as contingencies that moderate the above relationship?

In our research, we thus bring innovative contributions to the literature through multiple avenues. As a groundbreaking work, this study addresses a significant gap in the literature by examining the essential role of inter-organizational capability, specifically group sustainability decision comprehensiveness, in SSCM performance. Primarily building on both OLC and its recently updated version termed LLE, this study offers theoretically grounded solutions and highlights the relevance of both theories in classifying the complex studied subject from a previously underexplored angle. While existing research shows mixed results regarding the general effects of decision comprehensiveness on performance, this study, leveraging LLE as a foundation, identifies a *J*-shaped effect particularly between group decision comprehensiveness and the two sustainability outcomes of economic and social performances. By incorporating contingent mechanisms in designing moderating solutions for specific SSCM scenarios, this study deepens our understanding, adds better insights, and widens the scope of the applications of our findings. Finally, a set of practical guidelines supported by empirical evidence is offered for designing decision-making comprehensiveness and developing effective strategies to carry out appropriate sustainable policies in the supply chain.

The remainder of the paper is organized as follows. Relevant literature and theories are reviewed, and hypotheses are formulated. This is followed by field research employing a mixed-method approach. The article concludes with a discussion of the findings, their implications, and avenues for future research.

2. Literature and hypotheses

2.1. Literature review and theoretical background

Decision-making comprehensiveness and organizational learning Decision-making comprehensiveness refers to the degree to which an organization strives for thoroughness and inclusivity in formulating and integrating its strategic decisions [13]. According to Curşeu and Schruijer [5], the comprehensiveness of decision-making at the supply chain level is achieved through diverse supply chain alliances engaging in group decision-making processes, leveraging various perspectives. The comprehensiveness of decision-making at an inter-organizational group level, such as the supply chain, emphasizes not only an understanding of the ambiguous information but also the complex interdependencies between the interests of both partners and the thoughtful selection of the most promising decision alternatives by alliance managers [22].

Business decision-making, especially regarding comprehensive decisions, has been asserted to directly influence the manner in which organizational learning unfolds [16]. The holistic approach of group decision-making comprehensiveness—including scanning, collecting, and interpreting surrounding data—is a cornerstone of organizational learning when harnessed correctly [15]. As scholars assert, decision-making comprehensiveness motivates decision-makers to acquire knowledge to identify and evaluate entrepreneurial opportunities, ensuring they steer clear of ignorance and doubt (e.g., [19]). The relationship between decision-making frequency and organizational learning is further elucidated by Eisenhardt [28], emphasizing that executives propel their learning curve through the act of making decisions. However, a lethargic decision-making process can stunt this learning, making it problematic for long-term growth and adaptability [ibid]. Additionally, Carmeli and Schaubroeck [29] emphasize organizational learning in the decision-making process by noting that organizations, through reflection on comprehensive decision-making, tend to acquire a higher resilience to adversity. This process not only empowers such organizations with the ability to effectively manage crises but also immerses them in the more profound meta-learning realm [ibid]. In conclusion, the comprehensiveness of group decision-making acts as a linchpin in fostering a positive learning environment within an organization.

Group decision-making comprehensiveness in SSCM While extensive research on the impact of decision-making comprehensiveness on performance exists, there is notably scant empirical research addressing group decision-making comprehensiveness at the inter-organizational level. As demonstrated in Table 1, one of the few existing research projects, Walter et al. [22], delves into this inter-organizational sphere, investigating the consequences of inter-organizational decision-making comprehensiveness on performance (albeit primarily focusing on general performance metrics). Sequentially, Zwikael and Meredith [23] expand upon this research landscape by providing an investigation into the realm of inter-organizational decision-making comprehensiveness, further assessing its impact on a more specialized facet of performance—investment performance.

In terms of group decision-making at an inter-organizational level in SSCM, the literature covering this topic encompasses several approaches to improving the quality/outcome of decision-making. For instance, Allaoui et al. [8] propose a framework for collaborative decision-making tailored for sustainable supply chain planning. Methods such as simulations, Fuzzy TOPSIS, linear programming models, and case studies by scholars such as Salmela and Huiskonen [30] have also been applied to construct decision-making models to aid in supply chain alliances' collaborative efforts surrounding the decision-making of sustainable supply chains. Despite the current body of related research, there remains a noteworthy literature gap in the understanding of how comprehensive levels of group decision-making translate into genuine impacts on SSCM performance.

Through a comprehensive review of the literature, a conspicuous deficiency in the existing body of knowledge is unveiled: a dearth of research exploring the effects of decision-making comprehensiveness on SSCM performance from an inter-organizational perspective. Table 1 exhibits a review of related extant scholarly works. As observed in Table 1, an exception refers to Curşeu and Schruijer [5], who put forth a conceptual framework discussing the issue without empirical evidence. This research endeavor therefore seeks to address and bridge this gap by formulating a theoretically grounded framework and subsequently conducting empirical investigations to ascertain the influence of group decision-making comprehensiveness on SSCM performance, particularly the three aspects of sustainability performance—economic, social, and environment—within the context of interorganizational dynamics.

Organizational learning curve and learning from limited experience in SSCM decision-making While supply chain management represents a complex decision-making challenge [31], theories of organizational learning have been integrally linked with various supply chain processes, such as lean management [32], to help better elucidate the knowledge accumulation process, thus improving decision-making quality and overall supply chain performance (e.g., [33]). To effectively incorporate sustainability within supply chains, it is imperative to embed the requisite sustainable guidelines within the organizational culture through learning mechanisms [34]. One particular early work [35] underscores the potential of organizational learning theories to considerably enrich the domain of sustainable development by bringing forth a dynamic viewpoint. As a result, different organizational learning theories and concepts have been applied in the context of SSCM (e.g., [33,36]). A prominent and traditional organizational learning theory utilized in SSCM is the "organizational learning curve" (OLC) (e.g., [ibid]). This theory posits that firms enhance organizational outcomes through accrued experience, notably, the most pronounced progress and diminishing rate of improvement made manifest during the initial and concluding phases of experience accumulation, respectively [25].

Despite the above, the strategic complexity of decision-making in SSCM is marked by complex dynamics, delayed feedback, the collaboration of numerous interconnected decision-makers etc. [31]. Moreover, decision-making in SSCM also encompasses considerable opportunity costs and prospective benefits while being influenced by both market and competitive factors [24]. As a result, a substantial amount of experience is imperative before decision outcomes consistently exhibit an upward trajectory [ibid]. Relying solely on the beneficial effects of accumulated experience thus fails to adequately capture the dynamic nature of decision-making within the SSCM context. For this reason, the LLE theory introduced by Musaji et al. in 2020 [ibid] provides an innovative lens and supplements traditional OLC to provide a thorough explanation of SSCM decision-making. LLE theory considers limited experience in the initial phases to better explain the relationship between accumulated experience and outcome performance in the practical context of strategic decision-making. LLE suggests that the shape of the relationship between accumulated experience and performance is critically determined by costs and benefits. In its initial phases, this relationship reveals a downward trend due to the outweighed costs caused by insufficient experience, while after a series of sense-making activities, trial-and-error, and knowledge accumulation, positive tendencies in performance will be obtained owing to the significant benefits gained from accumulated experience. Given these traits, the LLE theory postulates that accumulated experience has a positive curvilinear effect on performance.

2.2. Conceptual model and hypothesis development

Decision comprehensiveness and sustainability performance In the SSCM context, group decision comprehensiveness describes the extent to which supply chain members achieve sustainable outcomes through an extensive decision-making process wherein multiple solutions are covered by search and analysis using quantitative and qualitative information to the greatest possible extent [5]. SSCM performance denotes the progressive sustainability achievement in SCM of a firm as demonstrated by its commitment to economic, social, and environmental sustainability measurements [37]. Economic sustainability denotes the attempt to reduce supply chain costs while balancing business interests with other strategic sustainability goals such as social and ecological aspects of the community, thereby enhancing its economic value [38]. Social sustainability signifies sustainable practices serving to solve social issues and add value to communities by increasing the human capital of individuals and furthering the societal capital of communities such as labor safety and social justice (see, [ibid]). Environmental sustainability refers to the reduction of environmental burdens such as waste disposal, pollution, greenhouse gas emissions, etc. [39].

The process of seeking out and analyzing relevant information for comprehensive decision-making in different aspects of SSCM can vary due to the availability of background information and domain knowledge in that sector, which in turn affects the manner of organizational learning. In particular, economic sustainability is typically misunderstood as only referring to business performance and profitability [40], while the emphasis should be on the balance of one's own interests with those of others. Current dramatic changes in the development model, demand structure, economic structural imbalance, and extreme constraints of energy resources, etc. [41] have made related information and knowledge about economic sustainability erratic, tacit, and difficult to acquire (cf., [42]). Similarly, social sustainability comprises actions exceeding organizational obligations to further cover other aspects [43]. For instance, OHSAS18001 provides guidelines for safety and health, while SA8000 offers guidance for workplace practices [37]. However, genuine social sustainability should move beyond these guidelines and integrate ethical norms [44] as well as employee well-being initiatives [45]. Furthermore, social sustainability contains variances across industrial fields with no established measurements in place [46]. Despite the growing proliferation of such research, the background information and domain knowledge of social sustainability are nonetheless underdeveloped and constantly evolving in step with social development.

In comparison to economic and social sustainability knowledge, due to increasing concerns over environmental sustainability, a considerable number of environmental laws and conduct codes have been developed and enacted in response to environmental deterioration [39]. As noted above, compared to its environmental counterpart, the acquisition of information and knowledge pertaining to economic sustainability is often challenging, given its erratic and tacit characteristics (cf., [42]). Also, the boundless benchmarks of social sustainability may deter supply chain partners from exceeding their obligations [47]. However, environmental laws and codes of conduct offer clarity, thereby promoting suppliers' receptivity and motivating their eagerness to learn [48]. In addition, numerous firms have been recognized as having introduced and conducted an extensive series of environmental sustainability supply chain practices [49], but such firms have failed to perform their social sustainability activities. For instance, Wal-Mart—renowned for its rigorous and forward-thinking environmental sustainability guidelines within its supply chain—has faced criticism concerning its treatment of individuals throughout that same chain [45]. Furthermore, while social sustainability presents challenges to mastery [50], implementing environmental sustainability practices necessitates less direct investment from the focal firm [36]. These evidence the widespread availability of environmental sustainability information/knowledge, particularly in the supply chain sector.

Owing to its tacit nature and the constant evolution inherent in economic and social sustainability knowledge, the quality of decision-making for augmenting economic and social sustainability performance is challenged by the depth of knowledge and the accumulation of experience in these domains. As supported by the LLE theory articulated by Musaji et al. [24], in the context of strategic decision-making, with its potentially unreliable learning and profound implications, knowledge and experience accumulation is important because it can be calibrated to improve performance. In the baseline case, a novice firm and its alliances base their sustainability decisions simply and primarily on instinct and heuristics, plus what are perceived to be the most decisive facts or impactful events, rather than through an extensive process [ibid]. As such, only a certain level of performance will be reached due to limited knowledge and experience in this subject. However, when the firm begins craving higher quality decisions through an increased level of comprehensiveness in decision-making processes, more systematic information collection and material processing, in addition to more intensive information exchange with alliances, are required (cf., [13]). Up-front investment is necessary for these activities [51]. Despite this, at this early stage of process comprehensiveness, the obligatory reliable and useful materials for better decision-making analyses may be insufficient or missing entirely due to the limited availability of economic/social sustainability information and knowledge, lack of understanding of the subject causality, and unaware of critical information/materials for assessment [24]. Hence, the ability of a firm to distill information/materials and exercise judgments is limited [ibid]. While the benefits of the initial investment may be marginally observed, such obstacles will encumber the firm and its alliances with significantly higher costs (e.g., capital and human involvement) to meet the requirements of the increasing comprehensiveness level, leading to drops in financial and economic value. Moreover, since the increasing demand in material collection and information exchange requires a higher level of competence (see, [20]), the insufficiency of relevant economic and social sustainability knowledge constrains the firm, necessitating better analysis of the gradually overloaded, complicated inflowing materials.

Notwithstanding, the cognitive burden of the overspill and consequent considerations may even distract a firm's attention, thereby leading to inaccurate inferences, inappropriate strategies, and disruptions in the operational execution of economic and social practices [52]. Moreover, this inexperience and inadequacy will heap pressure on internal stakeholders such as mid-range managers and front-end employees [53], increasing their perception of the instability/uncertainty of sustainability implementations. As internal stakeholders (especially front-end employees) are the firm's link to customers and the public, external communication efficiency may be jeopardized, and its social image will be undeservedly damaged. Eventually, these factors hinder social sustainability performance (see, [37]). According to Musaji et al. [24], suffering due to a lack of knowledge and experience suggests that economic and social sustainability performance will continue to decline until sufficient knowledge and experience are accumulated just to eventually break even.

After surpassing the threshold point, the firm and its alliances will gain positive momentum in performance enhancement from this accumulative knowledge—now the firm's distinct advantage—through collecting and analyzing substantial information [19]. A more extensive set of possible sustainability-practice alternatives is gained, and the underscoring benefits of firms' investments become more salient within their supply chain system. Through constant data assessment, the firm is more capable of gathering, disseminating, and synthesizing information and data [54]. According to the LLE theory, the combination of more alternatives and a higher aptitude toward managing materials enables the firm and its alliances to better evaluate the market in adapting their strategies to the changing environment [55]. Thus, the firm can find more appropriate sustainability strategies and implement them more efficiently, while its sustainability performance will be ensured to continuously improve upon the initial novice stage.

Given the above theoretical factors, sustainability decision comprehensiveness among supply chain alliances is therefore suggested to have a *J*-shaped relationship with economic and social sustainability performances. That is, the curvilinear effect initially trends negative, after which an upward trend of the curve will be observed, eventually outperforming the initial novice baseline point. H1 and H2 are thus formulated:

H1. A *J*-shaped relationship exists between sustainability decision comprehensiveness and economic sustainability performance.

H2. A *J*-shaped relationship exists between sustainability decision comprehensiveness and social sustainability performance.

Due to it typically being guided and regulated by governments [39] as well as having been practiced for decades [49], knowledge associated with those factors, such as conduct codes and government policies/laws toward environmental sustainability, are more mature and better understood than those for economic and social sustainability [56]. Seeking out the relevant information of environmental guidelines, which are indeed much more transparent (see, [49]), is simpler and less demanding. Firms are better equipped at collecting, managing, and further synthesizing required environmental information for better design of environmental policies, strategies, and tactics. As distinct from the other two aspects of sustainability, which commence at baseline novice status, this research draws on the OLC model [25], proposing that the environmental performance of a firm will always improve when experience is accumulated through comprehensive decision-making processes. In contrast, only a positive relationship with environmental sustainability performance is advised. Thus, we hypothesize:

H3. A positive relationship exists between sustainability decision comprehensiveness and environmental sustainability performance.

Moderating effect of competitive intensity Competitive intensity denotes a situation of rivalry among firms operating in an industry as typically characterized by the activities of competing firms—price competition, promotion competition, etc. [57]. Firms in a highly competitive intensity environment must seek out specific strengths to pursue differentiation [26]. Consumers then have more alternatives and can demand more from firms [57], such as lower prices and a greater variety of products/services with sustainability-related attributes.

As previously noted, with a lack of background knowledge and domain skills, plus high initial investment, group sustainability decision comprehensiveness increasing from low to moderate levels negatively impacts economic and social sustainability performance. The effect might be even more evident among firms with highly competitive intensity, as they will proactively devote greater effort to better understanding market opportunities and innovation, thus enhancing their knowledge and competence to differentiate themselves from competitors [26] and increasing their odds of survival. However, the heavy devotion to dealing with highly competitive intensity along with insufficient sustainability knowledge and data processing skills compounds the negative effects of sustainability decision comprehensiveness, jeopardizing economic and social sustainability performance for novice firms. However, at lower levels of competitive intensity, motivations are low for enhancing knowledge and signaling the sustainability of firms' offerings [58]. These firms will pay less attention to the scarcity of sustainability knowledge and associated skills, along with the investment in these aspects, thereby mitigating the adverse impacts of sustainability decisions.

As for moderate-to-high levels of group sustainability decision comprehensiveness, we concluded earlier that the knowledge and skills accumulated through the comprehensive decision-making process are key to boosting economic and social sustainability performance. We also expect that such positive effects might be more prominent among firms with high levels of competitive intensity. As denoted above, firms are motivated by necessity, pouring in more resources to innovate and differentiate themselves from competitors in such environments [26]. The elusive accumulated market knowledge and domain skills associated with economic and social sustainability, plus enhanced data processing ability, will equip firms with distinct strengths (see, [59]). Firms can then rapidly attain deeper insight, interpreting a broad spectrum of information to develop effective responses to marketing opportunities and problems, as such, distinguishing themselves from rivals and achieving higher profits. Positive impacts at subsequent stages will be magnified under conditions of highly competitive intensity. In contrast, at lower levels of competitive intensity, firms secure their performance regardless of their ability to satisfy their customers [60]. The acquired knowledge and skills become, in effect, less crucial, and any positive effects of the process are less evident. In sum, we postulate:

H4a. The *J*-curve relationship between sustainability decision comprehensiveness and economic sustainability performance is more pronounced among firms with higher, compared to lower, levels of competitive intensity.

H4b. The *J*-curve relationship between sustainability decision comprehensiveness and social sustainability performance is more pronounced among firms with higher, compared to lower, levels of competitive intensity.

For the positive linear relationship between group sustainability decision comprehensiveness and environmental sustainability performance, in a high-intensity competition environment, rather than accumulating readily available knowledge [61], differentiation is required and engagement in superior knowledge and creative responses is necessary. Thus, a superior competitive advantage can be sustained (see, [26]) as firms are driven to seek out new ideas and market information that are constantly changing and unfamiliar to competitors [59]. Compared to economic and social sustainability, environmental knowledge, such as that of green conduct codes and governments' ecological policies/laws, which is more widely acknowledged and adopted by numerous firms [49], will be less beneficial to firms in developing differentiation strategies. Therefore, firms will not effectively leverage environmental sustainability knowledge, which in turn dulls the positive effect of sustainability decision comprehensiveness on environmental sustainability performance for firms with high rather than low levels of competitive intensity. Hence, it follows that:

H4c. Competitive intensity negatively moderates the linear relationship between sustainability decision comprehensiveness and environmental sustainability performance.

Moderating effect of opportunism Opportunism refers to self-interest seeking with guile [[62], p. 56] and can be characterized by a wide range of behaviors, such as breaching contracts, obfuscating issues, shirking obligations, withholding or distorting information, and so on [27].

At low-to-moderate levels of sustainability decision comprehensiveness, firms operating in high levels of opportunistic environments are likely to experience a more noticeable negative effect on sustainability. Firms facing a high opportunism context tend to have asymmetric information problems in their partnerships due to unequal levels of expertise [63] and the information hiding behavior of their partners [64]. Therefore, firms will crave more relevant knowledge and skills in proactively taking measures to curb such opportunistic behaviors and avoid subsequent losses [ibid]. Additional resources will be allocated to address the shortage of economic and social sustainability information/knowledge and data processing skills and to deal with highly opportunistic behavior. However, in line with H1 and H2, at these initial and earlier stages, the greater the investment, the weaker the performance. Therefore, more devotion will intensify the negative effects of sustainability decision comprehensiveness on sustainability performance. In contrast, for low levels of opportunism, the collaboration and norms of trust built through supply chain strategic relationships will encourage the sharing of critical assets and information (see, [65]). In having less devotion, firms can benefit from their partners' resources, mitigating their lack of sustainable knowledge and information processing ability. This, therefore, redeems the negative effects of sustainability decision comprehensiveness on economic and social sustainability performance.

At moderate-to-high levels of sustainability decision comprehensiveness, firms with high, compared to low, levels of opportunism, are again anticipated to experience more robust growth in sustainability performance. Firms in environments with high opportunism levels tend to be skeptical of partners, cautiously establishing thorough control mechanisms [66]. To develop safeguards against opportunism, firms will devote resources to better predict potential scenarios for agreement term-and-condition development [67]. The accumulation of such understanding and knowledge related to economic and social sustainability, which are constantly changing and emerging, will become a unique asset benefitting the development of protection mechanisms when opportunism prevails (see, [68]). Hence, the positive effects of sustainability decision comprehensiveness on economic and social sustainability performance will grow, while the motivation to understand and evaluate partners' behavioral deviations is less evident at low levels of opportunism [69]. Hence, firms will deprioritize caution in monitoring and managing partner behavior (see, [66]). As the effects of decision comprehensiveness on sustainability performance become less significant, such positive effects are weakened. Therefore, we propose the following hypotheses:

H5a. The *J*-curve relationship between sustainability decision comprehensiveness and economic sustainability performance is more pronounced among firms with higher, compared to lower, levels of opportunism.

H5b. The *J*-curve relationship between sustainability decision comprehensiveness and social sustainability performance is more pronounced among firms with higher, compared to lower, levels of opportunism.

In the case of the positive linear effect of group sustainability decision comprehensiveness on environmental sustainability performance, at higher levels of opportunism, firms have a greater obligation to protect themselves from any negative consequences imposed on them by their counterparts [70]. Firms desire to obtain more relevant knowledge for anticipating upcoming challenges and implementing effective safeguards [67]. As such, tacit and readily available market knowledge tends to be more important for firms in mitigating high opportunism [68] in contrast to environmental knowledge. On these grounds, the enormity of the effect of sustainability decision comprehensiveness on environmental sustainability performance will diminish with higher levels of opportunism. So, we formulate:

H5c. Opportunism negatively moderates the linear relationship between sustainability decision comprehensiveness and environmental sustainability performance.

Following Gazzoli et al. [71], Shumanov et al. [72], and Wang et al. [73], a mixed-method approach comprised of two sequential studies—a quantitative survey and a subsequent qualitative study—is applied to test and validate the proposed hypotheses.

3. Study 1: a quantitative research

3.1. Measurements and questionnaire design

Since the focal studied constructs were not available from other data sources, a cross-sectional questionnaire survey was conducted to test the hypotheses following prior organizational learning literature, such as in the case of Zhu et al. [74]. Construct measures in the questionnaire were adopted using existing scales, with some modifications, to fit the research context under a seven-point Likert scale. A list of scales will be provided upon request.

The antecedent, group sustainability decision comprehensiveness, was operationalized based on the three-item scale of Simons et al. [12]. The outcome variables, namely the performances of the sustainable SCM—economic, social, and environmental—were assessed using the five-item scale drawn from Closs et al.'s [38] study, the four-item scale modified from Paulraj [75], and the six-item scale adapted from Blome et al. [76], respectively. For the two moderators, competitive intensity and opportunism were assessed via the four-item scales derived from Jaworski and Kohli [57] and Murtha et al. [21], respectively. Three control measures were also incorporated to partial out extraneous sources of variation in the outcome variables. First, information processing capability—a firm's ability to deal with loads of SCM information for the uncertainty mitigation purpose [77]—was measured using Hou et al.'s [78] four-item scale. Moreover, a firm's profile, including size (number of employees) and type (ownership: state-owned, foreign-owned, or private) were also comprised as control variables. All these variables were shown to correlate with firm performance in early research projects (e.g., [79]).

Since Vietnam is a non-English speaking nation and the original measurements were operationalized based on English literature, a double back-translation was carried out. The questionnaire was pretested with five practitioners having more than ten years of experience with SCM to ensure the quality of the survey in terms of readability, clarity, and appropriateness. Based on their feedback, a few changes were made to the wordings of some items and the format of the questionnaire, accordingly.

3.2. Quantitative data collection

The participating firms in the quantitative study (a methodology well suited to capture executives' minds and identify firms' practices and policies [80]) were manufacturers from a wide cross-section of industries operating in Vietnam. Abdul-Rashid et al. [81] demonstrated the essential role of sustainability in manufacturing industries due to common manufacturing problems such as enormous amounts of waste disposal, environmental damage, and energy overconsumption. Vietnam is an important manufacturing hotspot in Asia with its large labor workforce, low labor costs, geographical location, and open-door policies (see, [82]). Numerous sustainability development plans in Vietnam since 2011 have illuminated the need to adhere to sustainability criteria in developing their operational strategies. Moreover, an increasing number of national manufacturers have integrated social/-environmental responsibilities into their operations [83].

In Vietnam, the Ho-Chi-Minh City (HCMC) Metropolitan Area accounts for 40% of the entire national GDP [84], while manufacturing firms in this metropolitan area are more familiar with sustainability than those in other areas (cf., [85]). Binh-Duong province is in this area and, with its 30 industrial zones, has attracted investment from more than 30 countries worldwide [ibid]. The sustainability concept has recently been discussed at length for Binh-Duong industrial policy development, encouraging local firms to integrate sustainability into their operations. Manufacturers in its industrial zones constituted a significant set of representative firms within the study context (cf., [86]), making them an ideal choice for data collection.

In particular, manufacturers located in VSIP and My Phuoc—two of the five biggest zones—were contacted, with 500 of them willing to participate in our research. Questionnaires regarding sustainability practices were hand-distributed to eligible representative managers from individual firms, as their jobs were related to manufacturing operations such as production, procurement, etc. The use of a single respondent has been the bane of common method bias (CMB hereinafter) [80] and may not be suitable for firm-level studies despite its common use in recent literature for operations management (e.g., [87]). In this study, these key informants had significant responsibility in shaping the scope of their operations and thus held vast amounts of related knowledge. According to Raffaelli and Glynn [86], based on their practice, they could provide fact-based information "*at the aggregate or organizational unit of analysis by reporting on group or organizational properties rather than personal attitudes and behaviors*", thereby rendering this research as less prone to CMB. Other procedural measures were also executed before and during data collection to enhance the rigor of our research and alleviate/minimize the threat of CMB [ibid], such as the use of established scales, reverse-coded scale items, separation of measurement, performance of a pilot study, confidentiality and anonymity, unrelated items in between comprehensiveness and performance scales, etc. Moreover, respondents were given privacy when completing the questionnaire to avoid social desirability bias.

Finally, 321 questionnaires qualified out of the 327 that had been returned; the other six were excluded due to incomplete responses. The usable response rate was therefore 64.2 percent and the statistical power of the collected sample was above Cohen's threshold of 0.80, indicating a qualified sample size for this research. A table of the demographics of the participating firms in Study 1 will be provided upon request.

3.3. Quantitative data analyses and results

3.3.1. Bias countermeasures

Sample selection bias To address bias, several remedies were applied during the data collection procedure, such as data collection from different locations and having no distinction among respondents. During data analysis, Heckman's two-stage model was conducted. In the first stage, firms' export experience was chosen as the binary outcome of Heckman's *probit* model, in which all control variables were added together with firms' capitals as the instrumental variable. At Stage II, the Inverse Mills Ratio (generated at Stage I) was integrated into the structural models as a covariate for the sample selection bias correction to predict the three outcomes of SSCM performance. The results revealed insignificant Inverse Mills Ratio coefficients in all regression models, which suggested that our study was not impacted by the problem of sample selection bias.

Non-response bias Two separate non-response assessment procedures were applied. In Procedure I, the focal studied constructs—decision comprehensiveness, environmental performance, and competitive intensity—and all their respective measurement items among the first ($n = 80$) and the last quarters ($n = 80$) of the sample (representing early and late respondent groups, respectively) were compared. The results indicated no significant mean difference among all the constructs and scale items. In Procedure II, we further compared the difference of these constructs' scale items among the two groups regarding firm age and size, while again no difference was found. The results demonstrated that non-response bias was not a concern in this study.

Common method bias As noted above, before and during data collection, several procedural steps were made to mitigate possible CMB. Also, according to Siemsen et al. [80], there was no need to be concerned about CMB regression models with nonlinear and moderating terms, as this research was designed to test complex *J*-shaped interaction patterns. However, the possible effects of CMB were still calculated in this statistical control procedure. Harman's single-factor tests were applied with both the EFA and CFA (exploratory/confirmatory factor analysis) approaches, then a *post hoc* analysis of Lindell and Whitney's marker-variable test [88], and finally collinearity assessment (i.e., VIF values). In the first test, the EFA test resulted in seven factors with eigenvalues above 1.0, explaining 69.66 percent of the total variance, while the first factor only explained 27.09 percent of the total variance. Moreover, the CFA results demonstrated unacceptable fit indices with $\chi^2/\text{df} = 8.111$, CFI = 0.457, and RMSEA = 0.15, which suggested that no single or general factor emerged. In the second procedure, the second smallest positive correlation among the manifest variables in the study (i.e., 0.022) was identified and used to compute the CMB-adjusted correlations between all the variables [ibid]. The outcomes demonstrated a small difference between the original correlations and their CMB-adjusted counterparts (Δr=0.006–0.023). A small chi-square difference between the original and the CMB-adjusted models ($\Delta\chi^2$=50.43, Δdf=0) was also found, signifying the insignificant model-fit deterioration of the CMB-adjusted model. Finally, the collinearity assessment was considered to detect CMB. All VIF values ranged from 1.087 to 1.627, far lower than the threshold of 3.0 (a table of the analysis will be provided upon request). All these tests led to the contention that our research was free of CMB.

3.3.2. Accuracy analysis

CFA (Confirmatory factor analysis) was applied to evaluate the accuracy of the multi-item measurement scales. The measurement model demonstrated an acceptable model fit with χ^2/d.f.=2.303, RMSEA=0.064, SRMR=0.0578, and CFI=0.909. The measurement scale reliability was then assessed via Cronbach's alpha coefficient, composite reliability (CR), and average variance extracted (AVE) with thresholds of 0.7, 0.7, and 0.5, respectively. As shown in Table 2, all values came in above their respective thresholds. For convergent validity, all the items were loaded onto their corresponding variables with factor loadings above 0.5, except for one item pertaining to environmental sustainability, prompting its removal. Finally, the discriminant validity test was conducted by evaluating three criteria: correlation values, AVE values, and HTMT (Heterotrait-Monotrait) ratios. The results in Table 3 showed all correlation values to be less than the conservative cutting value of 0.8, while all constructs' shared variances with other constructs were lower than the square root of the AVE value, and all HTMT ratios were below the threshold of 0.9, indicating a qualified measurement accuracy.

3.3.3. Endogeneity

To eliminate possible endogeneity bias, we carried out the Durbin-Wu-Hausman test. Sustainability decision comprehensiveness was first regressed on all control variables and two instrumental variables (i.e., firm age and firm capital). It followed that the generated residual of this regression was incorporated as an additional regressor in the study's

Table 2
Measurement accuracy assessment.

Constructs	no. of scale items		Mean[a]/SD	α	CR[b]	AVE[c]	Item loading/highest cross loading
	Original	Final					
Competitive Intensity	4	4	4.93/1.02	.77	.79	.51	.71;.52; 0.84; 0.70
Opportunism	4	4	3.26/1.36	.83	.82	.54	.70; 0.91; 0.72; 0.57
Sustainability Decision Comprehensiveness	3	3	4.66/1.33	.83	.86	.66	.81; 0.76; 0.87
Economic Sustainability	5	5	5.45/0.97	.82	.85	.54	.70; 0.70; 0.91; 0.62; 0.72
Social Sustainability	4	4	5.60/0.93	.87	.88	.59	.91; 0.76; 0.83; 0.72
Environmental Sustainability	6	5	5.00/1.20	.85	.86	.56	.76; 0.81; 0.81; 0.69; 0.67
Information Processing Capability	4	4	5.15/1.22	.85	.86	.60	.78; 0.82; 0.76; 0.74

[a] Based on a seven-point Likert scale;.
[b] Composite Reliability;.
[c] Average Variance Extracted.

Table 3
Measurement accuracy analysis—Discriminant validity assessment.

Construct scales	CoI	Opp	SDC	EcS	SoS	EnS	IPC
Competitive Intensity (CoI)	**.714**	.285	.463	.300	.255	.306	.311
Opportunism (Opp)	.266	**.735**	.104	.141	.136	.124	.122
Sustainability Decision Comprehensiveness (SDC)	.467	.056	**.812**	.317	.432	.720	.719
Economic Sustainability (EcS)	.306	−0.099	.274	**.735**	.751	.522	.404
Social Sustainability (SoS)	.247	−0.108	.371	.669	**.768**	.554	.525
Environmental Sustainability (EnS)	.298	−0.014	.672	.500	.556	**.748**	.487
Information Processing Capability (IPC)	.310	−0.079	.639	.404	.507	.499	**.775**

Bivariate correlations and HTMT (Heterotrait-Monotrait) ratios are at the lower and upper parts of the diagonal, respectively, while diagonal elements are the square root of AVE (highlighted in bold).

hypothesized equations. The results indicated no statistical significance for the residual in all hypothesized regression models, suggesting that sustainability decision comprehensiveness was not endogenous in our setting.

3.3.4. *Hypothesized direct- and moderating-effect testing*

The processes proposed by Lam et al. [89] and Haans et al. [90] were implemented to test the direct and moderating effects. All examined predictors were mean-centered to eliminate potential multi-collinearity problems. The research framework with its proposed hypotheses is shown in Fig. 1, while the results are presented in Table 4.

J-shaped direct effects (H1-H2) Following Lam et al.'s [89] approach, we carried out hierarchical multiple-regressions in SPSS for *J*-shaped H1 and H2. We first tested their U-shaped relationships. As seen in Model 2, both the quadratic effects of sustainability decision comprehensiveness on economic (H1) and social sustainability (H2) were positively supported with β=0.20 ($p < 0.001$) and ΔR^2=0.04 ($p < 0.001$), and β=0.17 ($p < 0.01$) and ΔR^2=0.03 ($p < 0.01$), respectively. Following Haans et al. [90], we then calculated the marginal effects of decision comprehensiveness at the low and high end of the range (X_L and X_H), as well as the location of the turning point. For H1, as expected, the slopes at X_L and X_H were significantly negative and positive, respectively (β=−0.39 and 0.55, $p < 0.05$ and <0.01, respectively), while the turning point at −0.94 was well-located within the data range; all of this supported the presence of a curvilinear, concave effect. Similarly, the test for H2 also revealed supported results for the quadratic effect, with the slopes at X_L and X_H being β=−0.31 and 0.48 ($p < 0.05$ and <0.01), respectively, and the turning point at −1.10 falling into the data range. Secondly, the endpoint of the convex relationship above the beginning was assessed [91]. As seen in the graph-plotting of the hypothesized relationships in Fig. 2A1-2A2, the right end points in both graphs were above their corresponding starting point on the left side. As such, the curvilinear relationships were predominantly *J*-shaped, supporting H1 and H2.

Linear direct effect (H3) As can be seen in Model 1, the linear effect of

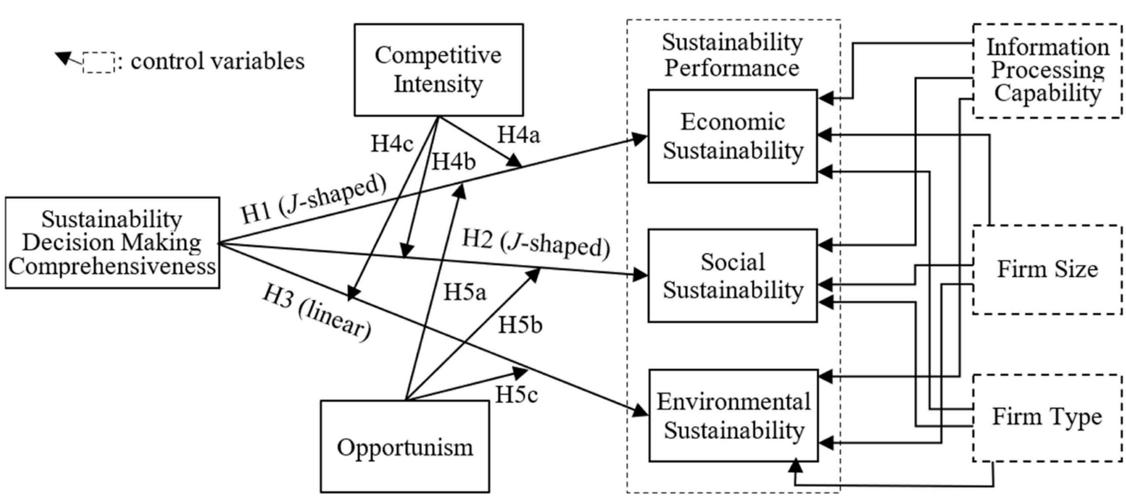

Fig. 1. Research model and proposed hypotheses.

Table 4
Hierarchical regression tests.

Variables	Model 1			Model 2		Model 3			Model 4		
	EcS	SoS	EnS	EcS	SoS	EcS	SoS	EnS	EcS	SoS	EnS
Control variables											
IPC	.28[c]	.36[c]	.15[a]	.31[c]	.38[c]	.35[c]	.42[c]	.15[b]	.30[c]	.37[c]	.15[a]
Firm size	−0.27[c]	−0.23[c]	.05	−0.25[c]	−0.22[c]	−0.22[c]	−0.20[c]	.05	−0.24[c]	−0.21[c]	.05
Firm type	−0.05	.02	−0.11[a]	−0.05	.02	−0.02	.03	−0.10[a]	−0.06	.01	−0.11[a]
Antecedents											
SDC	.21[b]	.22[b]	.50[c]	.21[b]	.22[b]	.20[a]	.17[a]	.50[c]	.25[c]	.25[c]	.50[c]
(SDC)^2				.20[c]	.17[b]	.27[c]	.24[c]		.14[b]	.12[a]	
Moderators											
CoI						−0.03	−0.08	.03			
Opp									−0.15[a]	−0.15[a]	.03
Interaction terms											
CoI*SDC						−0.26[c]	−0.21[c]	−0.10[a]			
CoI*(SDC)^2						.23[b]	.16[a]				
Opp*SDC									−0.11[a]	−0.10[*]	−0.06
Opp*(SDC)^2									.21[b]	.20[b]	
R-squared	.19	.24	.36	.23	.27	.33	.33	.37	.26	.30	.37
DR-squared				.04[c]	.03[b]	.10[c]	.05[c]	.01[a]	.03[b]	.03[b]	.00
Adj.R-squared	.18	.23	.35	.22	.26	.31	.31	.36	.24	.28	.36

IPC: Information Processing Capability; SDC: Sustainability Decision Comprehensiveness; CoI: Competitive Intensity; Opp: Opportunism; EcS: Economic Sustainability; SoS: Social Sustainability.
[a] < 0.05,.
[b] < 0.01;.
[c] < 0.001;.
[*] <0.1.

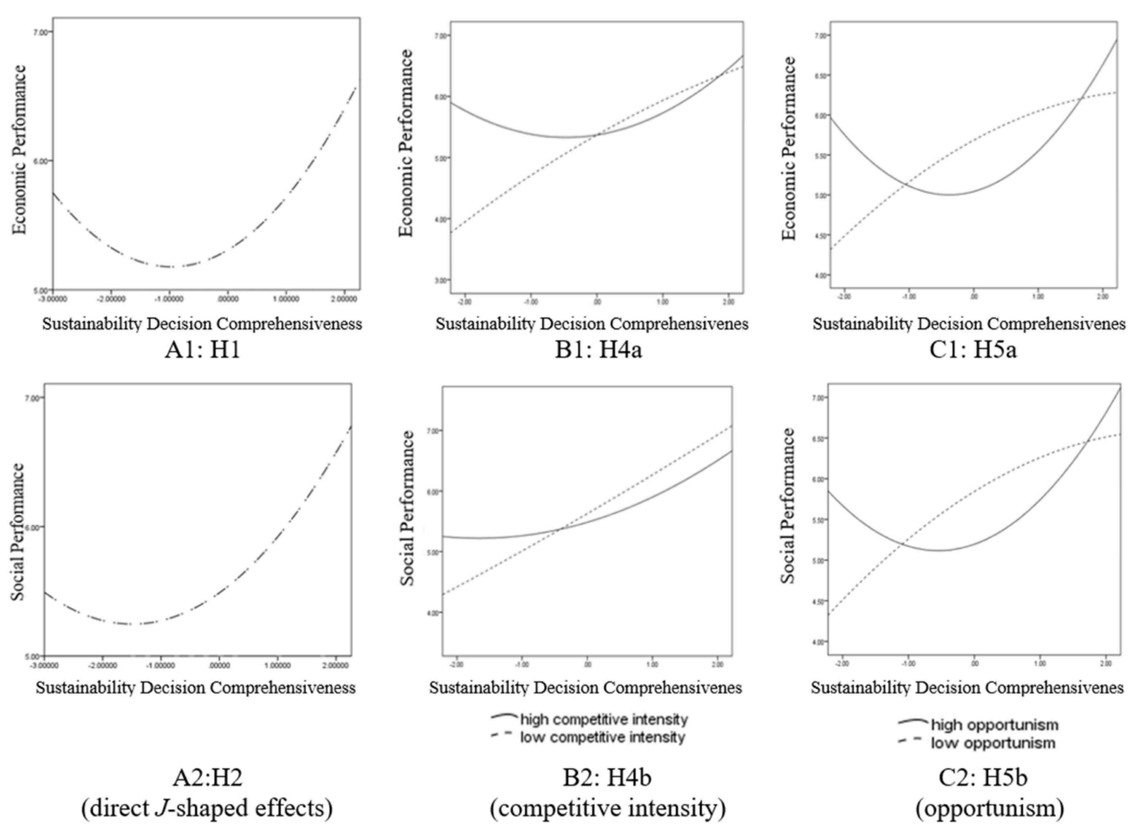

Fig. 2. Analyses of the *J*-Shaped effects of sustainability decision comprehensiveness on economic and social performances, and their corresponding moderating effects.

sustainability decision comprehensiveness on environmental sustainability (H3) was significant with its β=0. 50 ($p < 0.001$). Therefore, H3 was supported.

Moderating effect testing (H4a-4c and H5a-H5c) For this testing, the linear and quadratic interaction terms of the independent and moderating variables (competitive intensity and opportunism) were added to the equations in Models 3 and 4 (see Table 4), respectively, to test their moderating effects. As shown in Model 3, both the quadratic interaction terms for H4a and H4b were significant (β=−0.23 and −0.16, $p < 0.001$ and <0.05, respectively), while the moderating impact for the linear relationship (H4c) was negatively significant with β=−0.10 ($p < 0.05$). In Model 4, it was found that the moderator significantly affected the two quadratic relationships in H5a and H5b (β=0.21 and 0.20, both $p < 0.01$), whereas the conditional influence in H5c was negative, but not supported (β=−0.06, $p = 0.19$). The results indicated that except for H5c, all other moderating hypotheses (H4a-4c and H5a-5b) were supported. Table 4 shows the full regression results for each model.

The moderating impacts on the two quadratic relationships (H1 and H2 relationships) were further explored via the interaction plots [92]. As shown in Fig. 2, two interaction plots were developed, one with a low and another one with a high level of the moderating variable defined as one standard-deviation below and above the mean, respectively. Consistent with H4a-H4b and H5a-H5b, the interaction plots suggested that in the low moderating variable situations, the curves became a positive linear/slightly negative curvilinear shape, while in the high moderating variable values, the graphs were positive quadratic-shaped (see Fig. 2B1-2B2 and 2C1-2C2). The interaction plots provided further evidence to support all the moderating effects.

4. Study 2: supplementary qualitative evidence using semi-structured interviews

While the field quantitative survey effectively identified causal relationships, additional semi-structured interviews provided critical insights into these mechanisms. This further effort helped minimize the limitations of the quantitative study and achieved a more robust analysis.

4.1. Qualitative data collection

In Study 2, the data was collected through in-depth semi-structured interviews with top management in a supply chain environment. The interviews were discovery-oriented, encouraging open and fluid conversations to ensure they were informative and perceptive. The interview questions were developed according to widely accepted ethnographic research practices, incorporating elements such as confidentiality and anonymity, rich and descriptive content, broad and open-ended inquiries, and participating experiences [93].

A purposive sampling method was applied to identify qualified participants. This research specifically targeted top management involved in supply chain-related tasks, such as supply chain managers, from the HCMC Metropolitan area, with many drawn from the same industrial zones as in Study 1, to validate the quantitative findings of the prior study. To ensure robust results, participants were intentionally selected from a diverse range of industries (e.g., confectionery, detergent and cleaning products, and apparel) and various positions within the supply chain (e.g., manufacturers, suppliers, logistics providers, and retailers), along with distinct firm sizes, types, and the professional roles of the interviewees. Twelve participants were interviewed, with each section lasting approximately 50–90 min and audio-recorded for accuracy. The recordings were subsequently transcribed. The sample consisted of four branded manufacturers, two logistics providers, four suppliers, one retailer, and one distributor. Of the firms interviewed, 45% were multinational while the remaining 55% were Vietnamese. A detailed list of the interviewees and their firms from Study 2 is available upon request.

4.2. Qualitative data analysis and trustworthiness checks

Data analysis began with initial interpretations. After the interviews were transcribed, the transcriptions were carefully reviewed to gain an overall understanding of the data, with analytical memos produced to aid in interpretation. Relevant excerpts for the five proposed hypotheses were then extracted and translated into English.

To ensure the validity of the collected data and enhance the rigor of our interpretations, we implemented methods of trustworthiness as outlined in the literature [94,95]. Credibility was established by incorporating a broad array of questions in the interviews, encouraging participants to reflect on diverse perspectives such as exploring alternatives and considering responses from others. They were prompted to provide more detailed responses to clarify and confirm their viewpoints. For transferability, as noted above, Study 2 included interviewees with varied backgrounds, such as CEOs, directors, and supply chain managers from disparate industries, firm types, and supply chain positions. Data collection continued until theoretical saturation was reached. To maintain dependability, the emerging findings were frequently reviewed and scrutinized by two research team members while in cases of incongruities, discussions continued until a consensus was achieved. To uphold confirmability, we diligently documented and tracked all aspects of our analysis, including monitoring the coding process, maintaining an audit trail, etc. “Thick descriptions” were also employed in reporting our findings, furnishing verbatim participant quotes to uphold transparency and authenticity.

4.3. Findings from qualitative data

General overview The collected qualitative data provided deeper insights to enhance our understanding of the phenomenon under exploration. Interviewees acknowledged the significance of sustainability in the supply chain, as driven by regulatory pressures, customer pressure, and market demands. As stated by a deputy director, “The Vietnamese government has extensive and detailed environmental regulations that directly impact our business. Specifically, firms in the industrial zone have faced strict management requirements, with new investment projects required to undergo third-party environmental assessments and submit annual reports on impacts like air and water quality, noise, and wastewater treatment,” while a supply chain manager remarked that “starting in 2024, we should commit to adhering to the new packaging recycling law, which mandates that approximately 15% of our packaging must be recycled.” Many interviewees also acknowledged the increasing customer preferences and market demand related to firms’ sustainability performance, such as “we must align with market trends, particularly the demands of end buyers for standards like sustainability and green practices,” as related by a CEO, and “sustainability has become a major trend, with every firm pushing it forwards in various ways. Firms are all embracing sustainability to promote their brands and win consumer favor,” as noted by a supply chain manager.

Interviewees also highlighted the significance and involvement of group decision comprehensiveness among supply chain members when implementing sustainability. For example, in the words of a logistics manager: “The project of reducing carbon footprint is impossible without the participation of supply chain members. It took us about three months to negotiate solutions and benefits with all suppliers and distributors, from the time the plan was introduced… Afterward, these partners spent an additional three months preparing and beginning to roll out the plan, bringing the total process to around six months.” Another illustration was collected from a supply chain manager who indicated, “Involving the focal firm, suppliers, and distributors/customers is critical when considering various criteria to increase load capacity… Aiming to reduce the number of shipments and, consequently, carbon footprint, factors such as road readiness, warehouse conditions, business size, order frequency, and the mindsets of distributors and customers must be considered… The proposal to increase load capacity could only be made once all these criteria were met and distributors/customers were fully prepared.” Also, it was denoted by a CEO that “a recent customer order with stringent sustainability requirements has necessitated changes in fuel sources and supplier qualifications. Consequently, having multiple discussions across the entire supply chain, including the focal firm, suppliers, and customers, is essential.”

Variations in supply chain alliances’ understanding of the dimensions/types of sustainability performance were also uncovered. When asked about sustainability, most interviewees initially associated it with environmental practices, likely due to the prominence of environmental issues in government policies. For instance, as explained by one CEO: "Emissions, wastewater, and system design must meet strict standards for approval, as government requirements are becoming increasingly stringent." A supply chain manager outlined that "we have implemented various sustainability initiatives, including efforts to reduce carbon emissions, conserve energy, and minimize water usage. Additionally, we have recently introduced a program to collect and recycle our packaging materials." However, when asked about economic and social sustainability practices, interviewees surprisingly appeared less familiar with a more diverse set of standards. Regarding economic sustainability, interviewees showed a mixed understanding. Some viewed economic sustainability in terms of balancing costs and profit, as illustrated by the following statement: "[Sustainability is] the daily task of managing costs effectively to maximize profit," expressed one CEO. A logistics manager thought that "economic sustainability is a macro-level issue that should be addressed by the government rather than individual firms." Regarding social sustainability, in one case, one CEO observed, "Social sustainability relates to employees' welfare and benefits, but it is more challenging because it involves working with people and requires greater flexibility to retain them (employees)…" One director manager mentioned that "social practices are not fully standardized. In Vietnam, compliance is mandatory, but safety standards vary depending on the clients. While safety is generally governed by common legal requirements, some large clients impose their own stringent safety standards. This variability is further influenced by local regulations, customer demands, and corporate policies, particularly in industries like steel where safety is given high priority."

Qualitative evidence for H1–H2 Interviewees consistently reported that the initial emphasis on comprehensive decision-making in sustainability projects led to a temporary drop in economic and social performances. This was due to the significant investments required for systematic information collection and material processing, along with the limited availability of and subsequent capability to manage relevant sustainability data. As signified by one HSSE (Health, Safety, Security, and Environmental) manager in his observations, "[To determine the solar energy system], the legal procedures and safety measures were discussed thoroughly, which was time-consuming and costly, even though the actual installation might not take long." Moreover, one CEO declared, "[To optimize our transportation], we aligned all stakeholders and reached an agreement to integrate and digitize operational processes for real-time updates on stock levels… [in particular to minimize the carbon footprint in transportation], we initially invested in facilities and technology, such as installing GPS on trucks to collect routing data…" One deputy director also pointed out, "[In preparing to introduce the amfori BSCI System—a health and safety standard], we spent about two years on activities such as investing in senior staff to manage sustainability initiatives… Early on, our team made mistakes in decisions due to a lack of knowledge and had to independently research and learn regulations and decrees." Their experience suggested that an initial investment in data collection and processing was crucial for making decisions on implementing sustainability projects. Despite the investment, the reduction in economic performance during the initial stage was evident by "the initial costs were so high that breaking even wasn't feasible and it would take time to recoup the investment," affirmed a HSSE manager. Also, "meeting the strict criteria [for new environmentally friendly materials] required extra investment in systems, personnel, and more structured workflows, which prevented shortcuts and inevitably led to higher costs," underscored a chairman through his experience.

Pertaining to social sustainability performance, when firms and their supply chain alliances initially strived for comprehensive sustainability decision-making, their social image—and consequently their subsequent performance—declined as well. A lack of knowledge and experience in communication efficiency, alongside factors noted above, such as substantial information collection, material processing, limited capability, etc., resulted in a negative outcome. For instance, a logistics manager asserted that "when implementing new safety policies [to benefit our delivery vehicle drivers], they complained about wearing protective gear, considering it a waste of time… They found the safety shoes uncomfortable and were reluctant to hand over the keys due to concerns about losing valuable items in the truck… Therefore, they strongly resisted our safety policies, [which damaged our social image]." A similar issue of resistance arose and negatively impacted the social perception of the firms when a CEO put welfare mechanisms into place for distributors by training them to manage their sales teams. He elaborated, "They believed we were just adding more work, not realizing that we were trying to help them improve their sales and management skills." Several interviewees also expressed concerns about the "need for a shift in mindset towards [social] sustainability." In the initial stage of involving alliance members and investing resources to finalize strategies for increasing stakeholder and community capital, some partners still viewed these efforts as additional pressure on staff. This perception, at its core, jeopardized and unfairly tarnished the reputation of their sustainable social practices, ultimately undermining their social sustainability performance.

Despite the declining outcomes, interviewees echoed the point that the knowledge and experience accumulated from earlier sustainability decision-making efforts eventually streamlined and improved the collective decision-making process at a later stage. It happened because firms and their supply chain counterparts had cultivated the required ability to "grasp the problem more quickly since the approach process and solutions were quite similar" or "gain insights from previous failure," said a CEO and a supply chain manager, respectively. Hence, both economic and social sustainability performances improved as firms identified more effective sustainability strategies and implemented them more efficiently. For example, as expressed by a HSSE manager, "The knowledge and experience gained from previous sustainability decisions also helped the [entire supply chain] team manage the budget better, [which reduced costs and improved our economic outcome]," and the investment in learning and collecting information decreased, as one deputy director conveyed: "After three years, we no longer needed the consultant [for the amfori BSCI—a health and safety standard], which saved us a large amount of money. Our firm and supply chain counterparts were able to replicate the decision-making process independently, while continuously improving it to a higher level." An improvement in social sustainability performance was also evident when the firm's social image and reputation were better recognized by customers and buyers. For instance, a CEO revealed, "After we met the [health and safety] standards required by an existing customer, we leveraged that achievement to showcase our social sustainability records to other clients. New customers with less stringent standards were highly impressed and satisfied with our level of social responsibility, while those with higher standards recognized our efforts and encouraged us to continue improving."

Qualitative evidence for H3 The interview results disclosed that relatively transparent environmental guidelines and regulations enabled firms to process required information for comprehensive decision-making more effectively, thereby enhancing their environmental sustainability performance. An example of a firm successfully applying an environmentally friendly material was that "[non-plastic] materials are well-established and have been adopted by many large firms. Consequently, when collecting information for its implementation decision [i.e., replacing plastic materials with non-plastic options], our firm, along with our suppliers and customers, encountered no difficulties—except ensuring that this new material aligned with customers' product specifications… [Although a considerable amount of money was spent on testing], once we obtained the [eco] material certification, our environmental sustainability image was enhanced and recognized by both current and new customers," claimed a quality director. According to other interviewees, supply chain sustainability decisions regarding environmental practices made within alliances were guided by established industrial standards, which in turn enhanced their environmental performance as well. As relayed by a CEO, “When building the warehouse, our firm had determined to adhere to the Leadership in Energy and Environmental Design (LEED) green building certification system. This approach necessitated the involvement of various stakeholders, ensuring that sustainability was carefully considered and integrated at every stage. With the LEED qualification, we can demonstrate our commitment toward environmental sustainability to customers.” Consequently, the improved comprehensiveness in sustainability decision-making was unlikely to adversely affect environmental performance, as it was supported by a growing understanding of environmental information.

Qualitative evidence for H4a-H4c Our interviews suggested that competitive intensity enhanced the effects of decision-making comprehensiveness on economic and social sustainability performances. Many interviewees agreed that “in a highly competitive market, firms will invest more in sustainability to gain a competitive edge and attract customers.” However, when there was a deficiency of knowledge and experience in the industrial community for consultation while firms strove for sustainability decisions, the initial costs would be amplified. An illustration of this was that “to compete with rivals, we (i.e., our supply chain team) ultimately determined to invest not only in product quality but also significantly in sustainability implementation [among the supply chain system]. However, due to a lack of knowledge and experience [we could gather in advance], the investment in a sustainably standardized factory exceeded our actual needs. We are now facing the challenge of addressing overcapacity, [which has rendered our initial investment less economically viable],” described a CEO. Also, another CEO who tried to implement welfare mechanisms for distributors voiced concerns: “The decision-making process [for implementing the welfare program] within our alliances required more time and effort in urban areas, where competition was more intense. Distributors in these regions sold not only our products but also those of our competitors. As a result, urban distributors showed greater resistance to the additional workload associated with the new welfare mechanism [we intended to impose], [further affecting our public perception].” However, those who had already accumulated experience from sustainability initiatives displayed substantial confidence in their economic and social sustainability performances, enabling them to compete more effectively against opponents. A deputy director articulated: “In this highly competitive market, many firms aim to use sustainability records as their unique selling proposition. [Working together with supply chains on sustainability projects, experienced] firms like us that have already obtained a globally recognized health and safety certificate [i.e., amfori BSCI] are better recognized [by communities] for their social sustainability. We have thus been prioritized on the list of suppliers and buyers, which has attracted new business and led to more orders, thus increasing our revenue.” The interviews brought to light that, under competitive intensity, the initial pursuit of comprehensive sustainability decision-making may strain a firm’s economic and social performances due to inexperience; yet accumulated sustainability experience ultimately yields greater benefits.

Despite, better comprehension and acknowledgment of environmental sustainability reduced the necessity for accumulating accessible knowledge. Instead, distinguishing oneself from rivals became crucial, with active involvement and innovative solutions gaining greater importance. Thus, in a market with significant competitive rivalry, although “firms will invest more in [environmental] sustainability,” however, such an investment to maintain a superior market position might be less feasible. Firms should explore and develop new insights into evolving ecological market intelligence. As posited by one CEO, “The use of eco-friendly materials has become easy to replicate. For example, in our industry, most firms have already promoted their products as environmentally safe and made from natural ingredients. While possessing environmental sustainability knowledge and experience, [merely increasing investment] without breakthrough environmental innovation will not truly differentiate us. [In such a case], it will be difficult to remain competitive in the market.” Therefore, intensive industrial pressure could blunt the beneficial impact of comprehensive sustainability decisions on environmental performance.

Qualitative evidence for H5a-H5c Evidence of opportunism could be observed in the process of comprehensive group decision-making during sustainability implementation. A CEO made a complaint that “suppliers attempted to violate agreements to advance their own interests.” A deputy director grumbled that “some sustainability requirements were not met by the suppliers, yet they misrepresented compliance during evaluations.” A logistics manager lamented that “when inspections were conducted to ensure CO_2 emission standards, logistics suppliers attempted to conceal non-compliant trucks.” Under these interviewed conditions of heightened opportunism, the impact of decision-making comprehensiveness on sustainability performance outcomes would become more significant. Specifically, these opportunistic behaviors of supply chain stakeholders made it more difficult to collect required information with higher costs and efforts needed to make comprehensive decisions at the early stage. That is, efforts to mitigate opportunistic behaviors tended to exacerbate the initial costs of comprehensive decision making; as such, the negative effects of decision-making comprehensiveness on both economic and social performances were amplified. For example, a CEO conveyed that, “The supplier withheld certain information from us, likely out of concern that we would impose extra requirements… [We] realized that more effort should be put into collecting extra information for making decisions, rather than solely relying on that supplier’s suggestions. However, information constraint led to emerging problems, resulting in unforeseen costs to resolve them and placing significant pressure on our staff, [thus impacting our economic and social sustainability outcomes].” Another instance underscored the rise in costs stemming from opportunism, as shared by one supply chain manager: “When soliciting bids from logistics providers, we requested qualified eco-trucks with low CO_2 emissions at reasonable costs. Nevertheless, the supplier who initially won the bid due to its low cost ultimately canceled the contract because the firm promised more than it could deliver. This caused significant financial loss and required considerable human resource efforts to recollect the information and carry out our supplier selection process among alliance partners [which jeopardized our economic sustainability performance and societal standing as we failed to fulfill our promise for eco-logistics practices].” However, those with experience in sustainability decision-making within their supply chain alliances and in handling opportunistic behavior showed no signs of concern. They even held greater confidence in their ability to successfully manage decisions and execute sustainability initiatives across their supply chain systems. The supply chain manager referenced above went on to add, “Now, after gaining a better understanding of eco-truck types and their operation, we have established a mechanism to make improved decisions [in selecting eco-truck providers for CO_2 emission reduction] within our alliance. What’s more, a set of penalties has been introduced to address this issue, ensuring that suppliers commit to our transportation requirements. As a result, we have reduced the costs and workload associated with the decision of selecting appropriate eco-truck providers.” Another interviewee, a quality director, articulated: “Despite the presence of opportunism, our firm and alliances, through leveraging our experience and knowledge in sustainability supplier selection decisions, have established a frequent and systematic auditing schedule across our supply chain. This enables us to effectively manage and ensure that our suppliers adhere to both safety and social standards. Furthermore, it also allows us to identify any opportunistic behaviors at an early stage, safeguarding both our profits and social reputation from potential risks.” That is, when opportunism was prevalent, pursuing initial sustainability decision-making with more alliance partners might further burden a firm’s economic and social sustainability performances, as it required additional resources to address opportunism. However, efforts to address asymmetric information in opportunism eventually transitioned into yielding advantages at a later stage.

Notwithstanding, the role of opportunism in the decision-making comprehensiveness and environmental sustainability outcome relationship might appear somewhat obscure. As elucidated by a quality director, "Our environmental performance is regularly assessed by the government [through standardized, transparent regulations], making it nearly impossible for supply chain members to withhold and conceal information from us. This transparency ensures that opportunistic behavior has minimal impact on our sustainability efforts or decision-making processes." The interview indicated that the firm was certain of the necessary information and would request supply chain counterparts to provide it. A supply chain manager also brought up that "While truck providers may fail to uphold their commitment, leading to financial losses [driving up our economic costs] and increased pressure on our team [hurting our social image], I would not say it will negatively impact our environmental performance. It will certainly cause delays in our CO_2 emission reduction plans within our alliances, but we can fix the issue and be quickly back on track to meet our sustainability targets." When supply chain alliances acted opportunistically, requiring the focal firm to invest additional efforts to address the issue, the interview revealed that there might be no difference in its impact on the effect of decision-making comprehensiveness and environmental sustainability performance.

5. Conclusion

This study examines the impact of sustainability decision-making comprehensiveness among supply chain alliances on sustainability supply chain management performance. First, the findings confirm the curvilinear *J*-shaped effects of sustainability decision comprehensiveness on economic and social sustainable supply chain performance, while the positive linear relationship of sustainability decision comprehensiveness and environmental sustainability performance is evidenced as well. Secondly, the tests on the moderating effects of competitive intensity and opportunism on the above two curvilinear relationships confirm our expectations in the projected moderating directions. More specifically, high levels of competitive intensity and opportunism are confirmed as creating more pronounced positive curvilinear effects, as predicted. On the other hand, at low levels of competitive intensity and opportunism, the two curvilinear effects are flatter in the case of competitive intensity and flip to a negative curve in the case of opportunism. Also, in terms of the moderating effect on the linear relationship of sustainability decision comprehensiveness on environmental sustainability performance, as hypothesized, the negative moderating effect of competitive intensity on this link is supported. However, the interaction of opportunism with this link is found to be insignificant, although still negative as proposed. This might be due to the popularity, transparency, and ease of collection of environmental knowledge, as well as the nature of the compulsory characteristics made by environmental laws and regulations (see [36,53]). These make it more difficult and less meaningful for partners to hide, twist, and opportunistically manipulate related knowledge and exaggerate their capabilities for their own benefit. Knowing this, firms might be still encouraged and motivated to pay, to some extent, certain levels of attention to ubiquitous environmental knowledge to enhance related performance. As such, in the case of high opportunism, the deteriorating phenomenon of the positive effect of sustainability decision comprehensiveness on environmental sustainability performance is undermined. The moderating effect is thus observed as insignificant, though still negative.

5.1. Theoretical implications

Our findings provide theoretical implications in several ways. Firstly, we advance research examining decision-making in the SSCM literature (e.g., [96]) by providing fresh insight into the ignored direct-effect relationship between group decision-making comprehensiveness and SSCM performance. Although prior research highlights the importance of decision-making in SSCM (e.g., [97]) and the comprehensiveness of decision-making has been observed to affect performance in the scope of internal firms (e.g., [19]), the critical role of sustainability decision-making comprehensiveness in SSCM performance is unclear. Therefore, we produce this empirical study as a pioneering work to examine such relationships. Secondly, we contribute to the theoretical richness of decision-making comprehensiveness in strategic management by clarifying a positive curvilinear, in particular *J*-shaped, relationship in this investigated relationship. While extant research related to the studied direct-effect relationship in general (i.e., not related to sustainability) yields mixed results (see, [17], for a review), our empirical study confirms the *J*-shaped effect of decision comprehensiveness on certain types of performance in the particular SSCM context. Thirdly, while there is a relative dearth of OLC theoretical application in SSCM, LLE—a recently developed theory revised from OLC [24]—has not been evidenced in the supply chain in general, and SSCM in particular. We ground our conceptualization through the OLC and LLE theories in the SSCM context. This empirical study reveals that while traditional OLC supports the explanation of the linear relationship between sustainability decision-making comprehensiveness and environmental sustainability performance, the LLE theory turns out to be an appropriate and novel theoretical background to understand the non-linear *J*-shaped effects of this important precursor on economic and social sustainability performance in SSCM. We, therefore, extend the proper theoretical applicability of both OLC and LLE in the complicated SSCM subject literature under study, but in a complementary rather than individual manner. Finally, while prior research about decision comprehensiveness emphasizes the conditional role of environmental uncertainty (e.g., [17]) in the relationship between decision comprehensiveness and performance in general, and more specifically the contingencies of competitive intensity and opportunism, it implies that the characteristics of environmental uncertainty have not been incorporated in the research. We contribute by enriching our knowledge of decision comprehensiveness by introducing relevant contingencies of competitive intensity and opportunism into our research model.

5.2. Managerial implications

This research also yields several valuable insights for managerial practices. In spite of the vital role and benefits of making quality SSCM decisions [97], the initial tradeoff between making comprehensive decisions and unguaranteed outcomes (see, [17]), especially while there is insufficient domain knowledge regarding economic and social sustainability, can demotivate firms to increase the comprehensiveness level of supply chain decision-making. In this light, our research provides convincing evidence of the outcomes of decision-making comprehensiveness efforts in the context of SSCM performance. Our findings indicate that although early financial and human resource investments may lead to adverse effects on social and economic sustainability performance at the initial stage, firms can expect prosperous results in the future. Furthermore, environmental investments for comprehensive decision-making are always helpful and beneficial for improving performance. Holistically speaking, the direct-effect results of this research suggest that ample, thoroughly considered sustainability decision-making produces better, more effective sustainability performance in the long run. Finally, our results reveal that when firms face high environmental uncertainty (i.e., competitive intensity and opportunism), higher preliminary costs are required to deal with these market challenges while simultaneously attaining decision-making capabilities for their supply chains. Therefore, it is harder to achieve better SSCM performance via sustainability decision-making comprehensiveness. However, we also demonstrate that continuous social and economic investments in enhancing sustainability decision-making comprehensiveness can be compensated for in the long term, rendering firms more competent in handling these highly uncertain conditions. In particular, an accumulation of social and economic knowledge and skills can enhance firms' competitive advantages, seizing upon opportunism and eventually standing out among rivals. However, when emphasizing social and economic sustainability performance under the contexts of intense competition and massive opportunism, there will be a tradeoff among the three aspects of sustainability performance owing to the declining environmental sustainability performance as decisions are made more comprehensively. Based on the above managerial implications, we advance firms' knowledge at the inter-organizational level, with a set of supply chain guidelines provided as to how sustainability decision-making comprehensiveness among supply chain alliances can be conducted in various contingencies.

5.3. Future research

As with other studies, our research is subject to certain limitations which provide promising avenues for future research. This study is conducted with a single informant, utilizing solely cross-sectional data. Despite being presented in a way that proactively addresses concerns about CMB, this research might still overlook a bias and introduce spurious decision comprehensiveness—sustainability performance relationships. If possible, it is advisable to undertake a longitudinal analysis incorporating three stages of surveys for the independent, moderating, and dependent variables, or opt for multi-source designs to more thoroughly alleviate potential CMV. While our research framework is based on the learning curve and its modified theory—i.e., OLC and LLE—other potential mechanisms can be determined to link sustainability decision-making comprehensiveness and SSCM performance. For instance, the sharing of knowledge is expected to occur during the decision-making process among supply chain members [5]. This can include the relevance and quality of shared knowledge, wherein members share unique knowledge, the breadth versus depth of shared knowledge, etc. These can be fruitful opportunities for future endeavors. In addition, this research recognizes the occurrence of shape-flipping curves in which the positive curvilinear shapes flip to negative curvilinear shapes due to the effects of moderators. According to Haans et al. [90], such a phenomenon provides interesting avenues if future research can interpret it in a theoretical and substantive way. Moreover, in OLC and LLE, knowledge and effort levels exhibit a critical role in the learning process and are associated with decision comprehensiveness. The potential thus exists for future research to delve deeper into these two concepts within the current study context. Moreover, firms' roles and power can be taken as another situational effect variable in the current research context. This research does not consider the roles and corresponding power of firms in supply chains. Firms' roles in their networks such as orchestrators and executors may influence their attitudes and behaviors (see, [98]). Future research, therefore, can introduce firms' roles and corresponding power in their supply chains as a contingent condition in the studied framework. Additionally, sustainability issues can be dynamic [81] while the current empirical study is based on a cross-sectional basis. Future research is recommended to use longitudinal types to capture the dynamics of the currently researched issue. The robustness of the research framework and findings would be confirmed if the analysis supports our results, while failure would suggest another direction for future explanatory studies. Finally, researchers can examine the proposed relationships, possibly bringing with them the growing availability of large datasets and big-data analytics. Complementing the assistance of artificial intelligence (AI) methods such as machine learning and deep learning, more sophisticated mathematical tools can be applied to analyze large databases, and new relationships may thus be explored and identified.

CRediT authorship contribution statement

Tessa Tien Nguyen: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Angelina Nhat Hanh Le:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Wesley J. Johnston:** Writing – review & editing, Supervision, Resources, Project administration, Methodology. **Julian Ming Sung Cheng:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

None.

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Supplementary materials

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Data availability

The data that has been used is confidential.

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